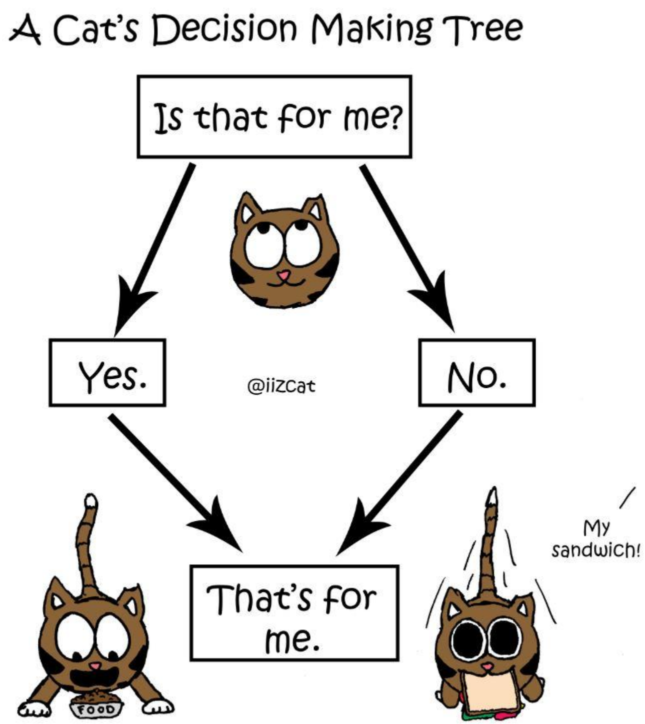
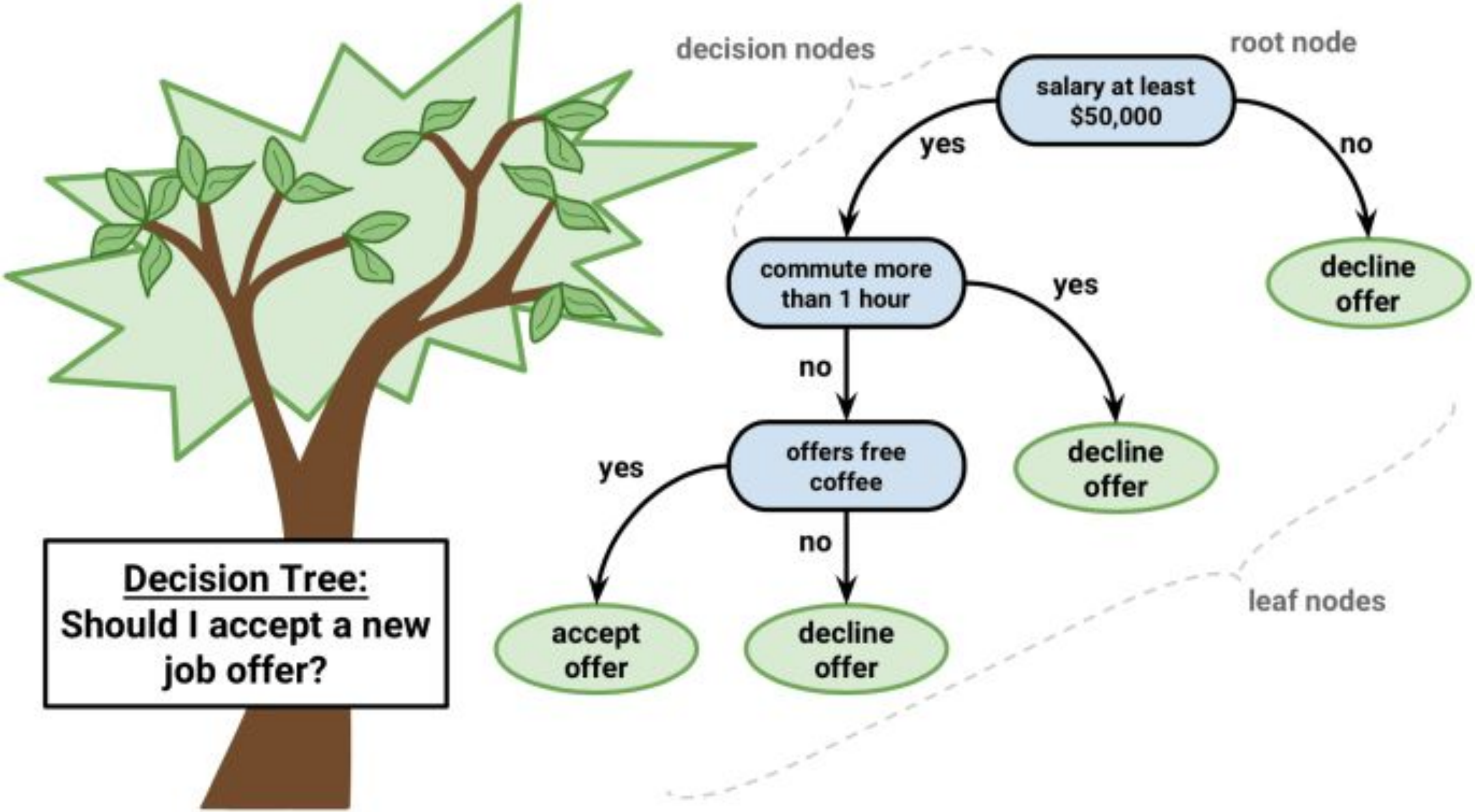
CSE 445: Machine Learning

Decision Trees

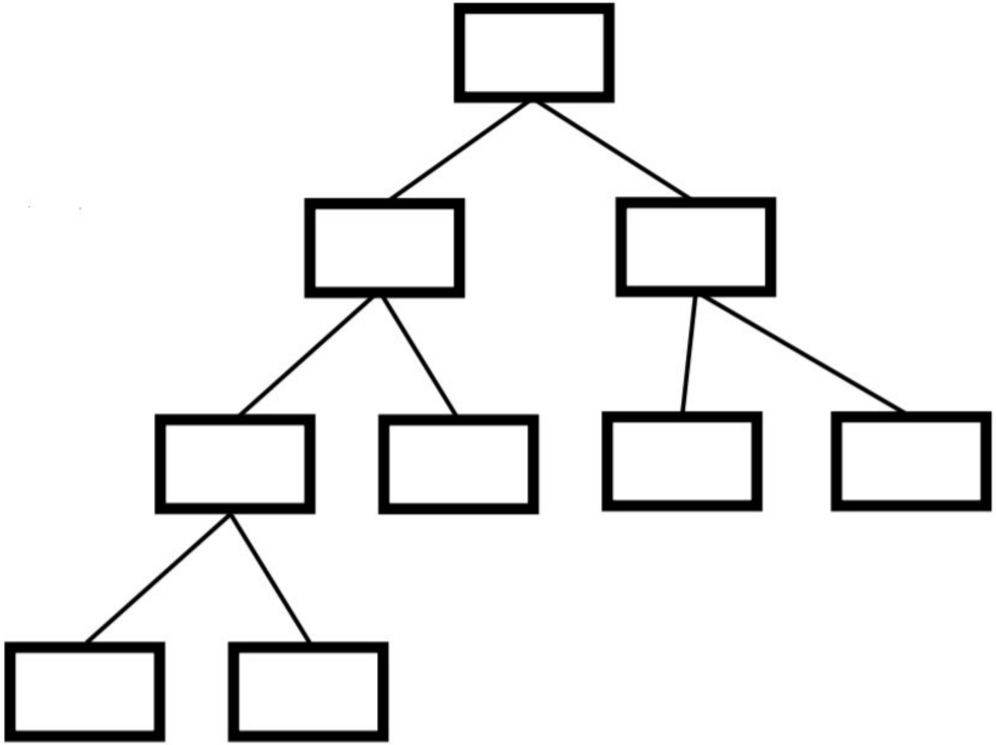


Decision Trees: Example



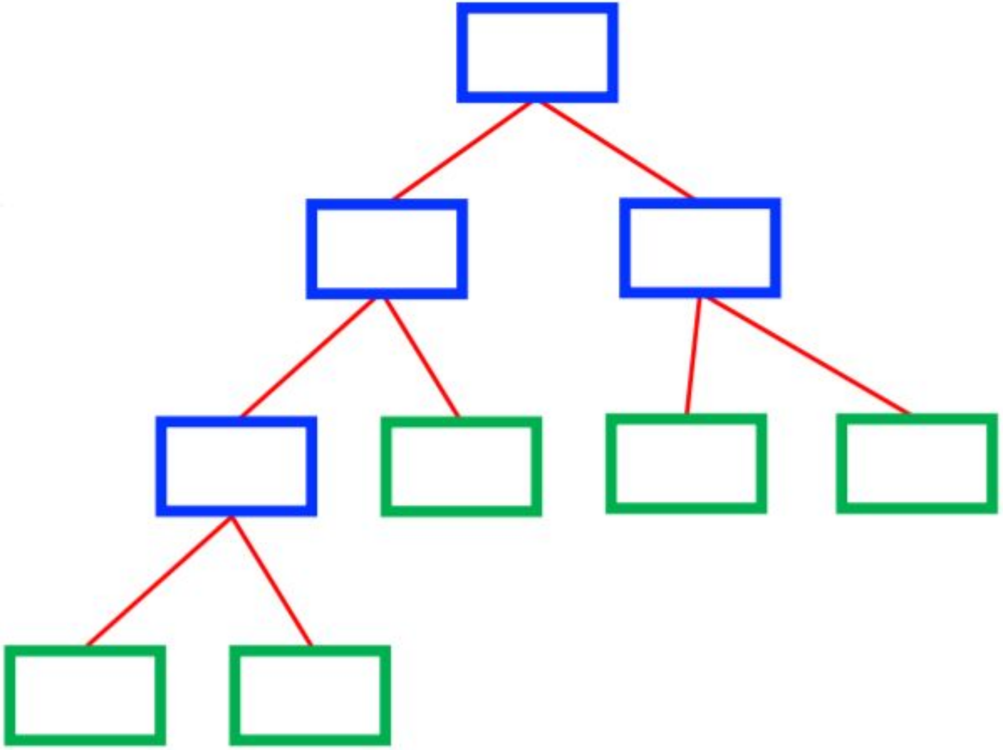
Decision Trees: Generic Structure

- A Decision Tree is a flow-chart-like tree structure
  - Internal nodes: test an attribute
  - Branches: test result
  - Leaf nodes: Class label
- At each node, one feature is picked to split training examples into distinct classes
- New samples are classified by following a matching path to a leaf node



Decision Trees: Generic Structure

- Goal: Predict Class label
  - Internal nodes: test an attribute
  - Branches: test result
  - Leaf nodes: Class label
- How to build(i.e. train) Decision Trees? Two major methods are CART or ID3
  - CART Classification and Regression Trees, Gini index used for best attribute
  - ID3 Iterative Dichotomiser 3, Entropy used for best attribute



How to learn a Decision Tree

- Greedy approach (NP complete problem)

Function BuildTree(m,A) //m:instances, A: attributes

if empty(A) or all m(L) are the same

status = leaf

class = most common class in m(L)

else

status = internal

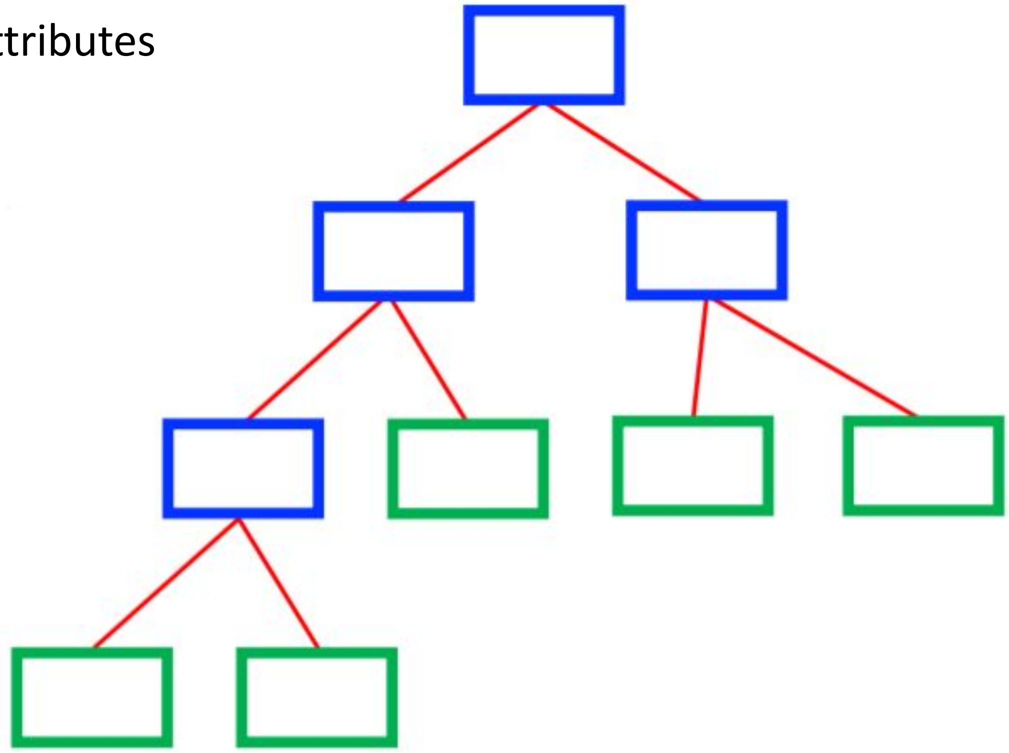
a □ bestAttribute(m,A)

LeftNode = BuildTree(m(a=1), A \ {a})

RightNode = BuildTree(m(a=0), A \ {a})

end

end



CART

- DT algorithm for producing binary classification or regression Trees
  - If target variable is categorial → classification, if numeric →regression
- Handles data in raw form without preprocessing
- Best attribute selected based on Gini impurity score

$$G_i = 1 - \sum_{k=1}^{n} (p_{i,k})^2$$

  - $p_{i,k}$ → ratio of class k instances among the training instances in i^{th}

- CART involves first splitting training set into two subsets using a single feature k and a threshold
- Then searching for the pair that produces the purest subsets
- CART classification cost function:

$$J(k, t_k) = \frac{m_{left}}{m} G_{left} + \frac{m_{right}}{m} G_{right}$$

- G_{left}/ G_{right} is Gini impurity score for left/right subset
- m_{left}/ m_{right} is number of instances in the subsets

CART Example

- Example problem: Predict if a person has heart disease or not based on three features using a DT

Chest Pain	Good Blood Circulation	Blocked Arteries	Heart Disease
NO	NO	NO	NO
YES	YES	YES	YES
YES	YES	NO	NO
...			

Sample dataset header

Attribute #1 : Chest Pain

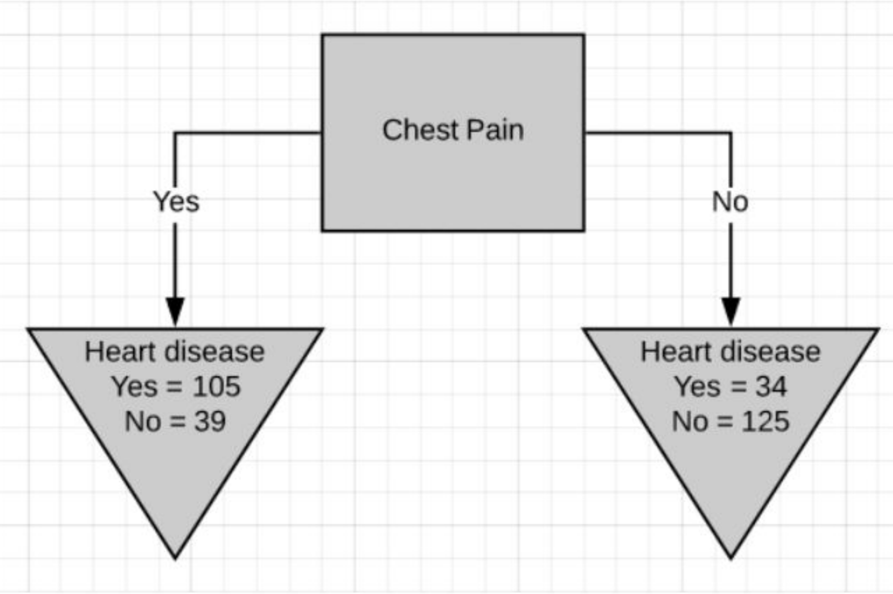


Fig.6-Chest pain as the root node

$$G_i = 1 - \sum_{k=1}^{n} (p_{i,k})^2$$
$$= 1 - (\Pr(Yes))^2 - (\Pr(No))^2$$

Left leaf Gini impurity

$$= 1 - \left(\frac{105}{105+39}\right)^2 - \left(\frac{39}{105+39}\right)^2 = 0.395$$

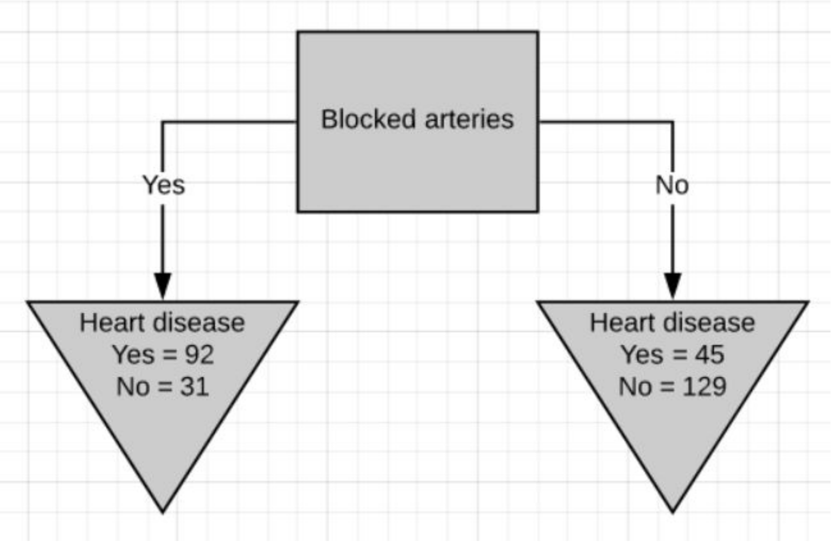
Right leaf Gini impurity

$$= 1 - \left(\frac{34}{125+34}\right)^2 - \left(\frac{34}{125+34}\right)^2 = 0.336$$

Total Gini impurity – weighted average

$$= \left(\frac{144}{144+159}\right) * 0.395 + \left(\frac{159}{144+159}\right) * 0.336$$
$$= 0.364$$

Attribute #2 : Blocked Arteries



Left leaf Gini impurity

$$= 1 - \left(\frac{92}{92+31}\right)^2 - \left(\frac{31}{92+31}\right)^2 = 0.377$$

Right leaf Gini impurity

$$= 1 - \left(\frac{45}{45+129}\right)^2 - \left(\frac{129}{45+129}\right)^2 = 0.383$$

Total Gini impurity – weighted average

$$= \left(\frac{123}{123+174}\right) * 0.377 + \left(\frac{159}{123+174}\right) * 0.383$$
$$= 0.381$$

Attribute #3: Good Blood Circulation

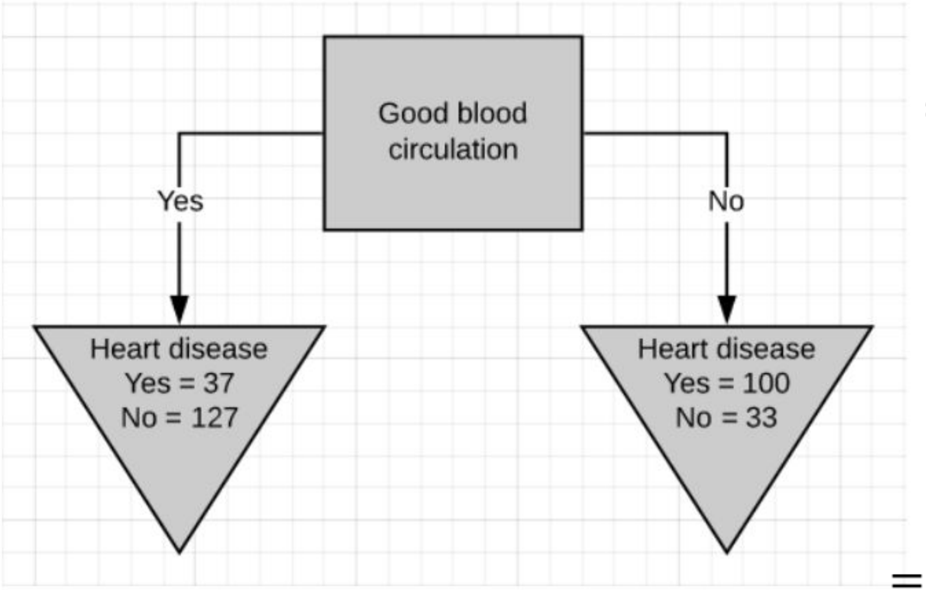


Fig.7-Good blood circulation as the root node

Left leaf Gini impurity

$$= 1 - \left(\frac{37}{37+127}\right)^2 - \left(\frac{127}{37+127}\right)^2 = 0.349$$

Right leaf Gini impurity

$$= 1 - \left(\frac{100}{100+33}\right)^2 - \left(\frac{33}{100+33}\right)^2 = 0.373$$

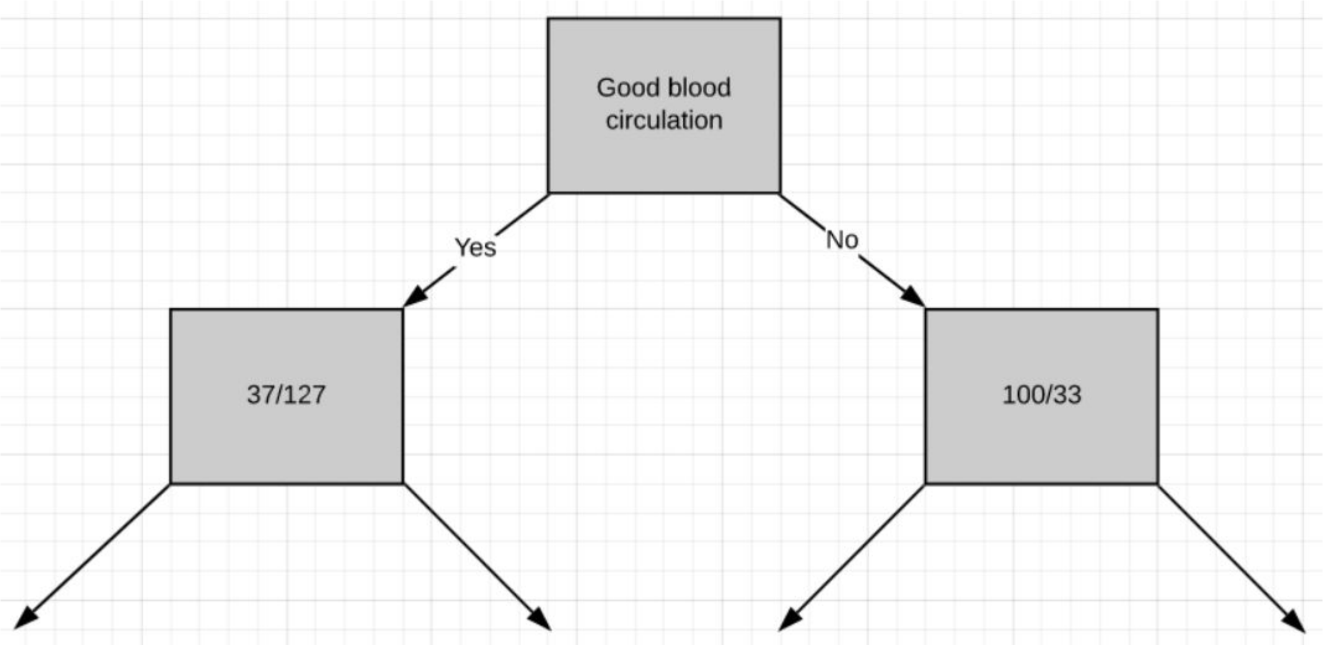
Total Gini impurity – weighted average

$$= \left(\frac{164}{164+133}\right) * 0.349 + \left(\frac{133}{164+133}\right) * 0.373$$
$$= 0.360$$

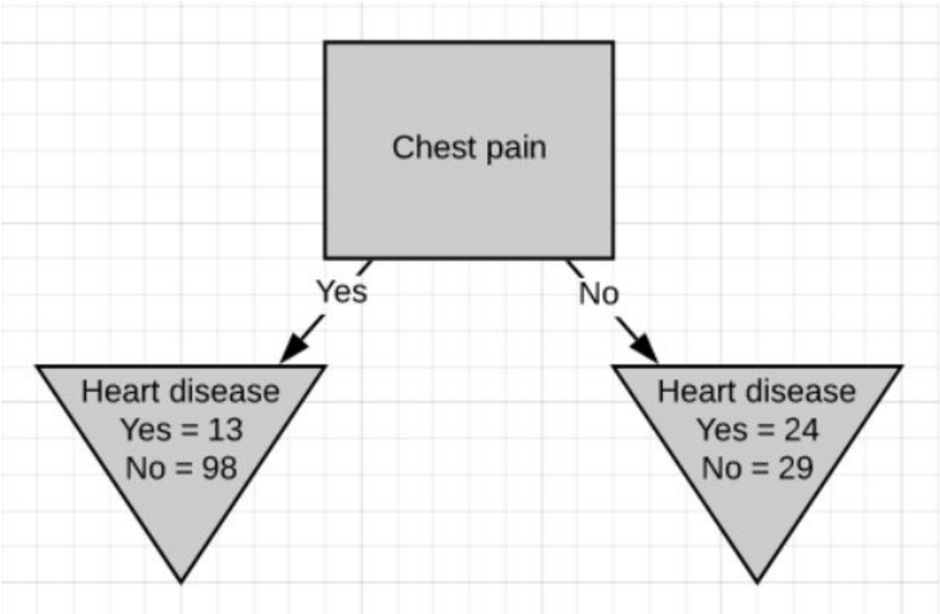
Lowest impurity with this attribute – set as root node

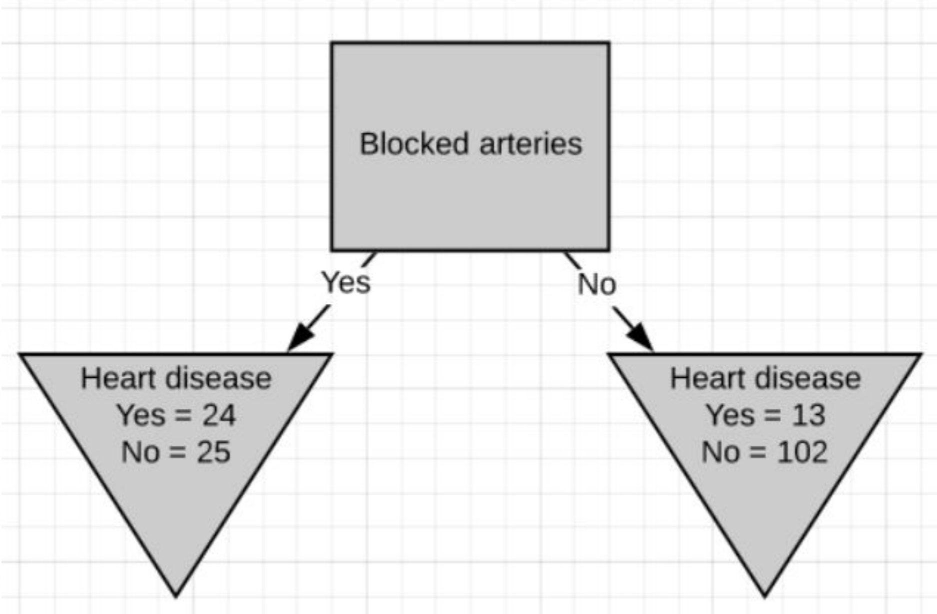
CART example

- Selecting GBC as root node divides the tree as follows
- Follow the same steps for the remaining internal nodes to find the decision tree

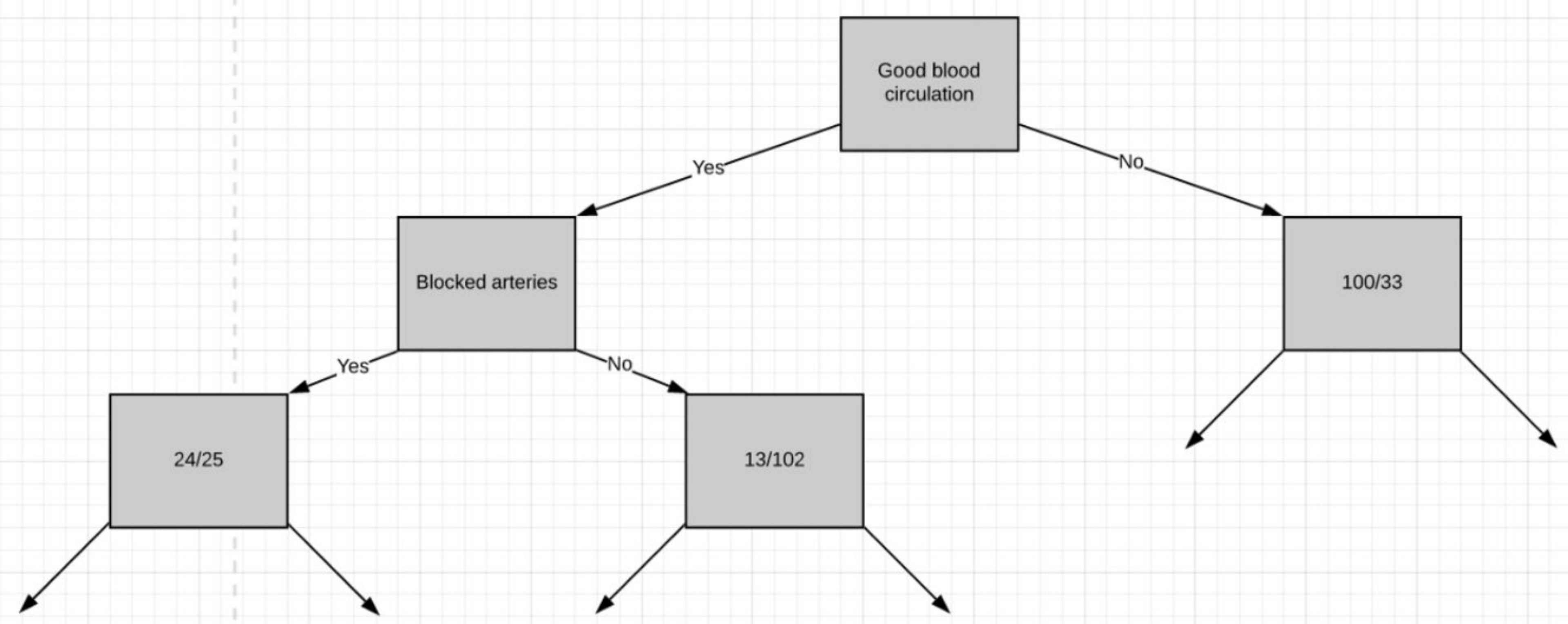


CART example

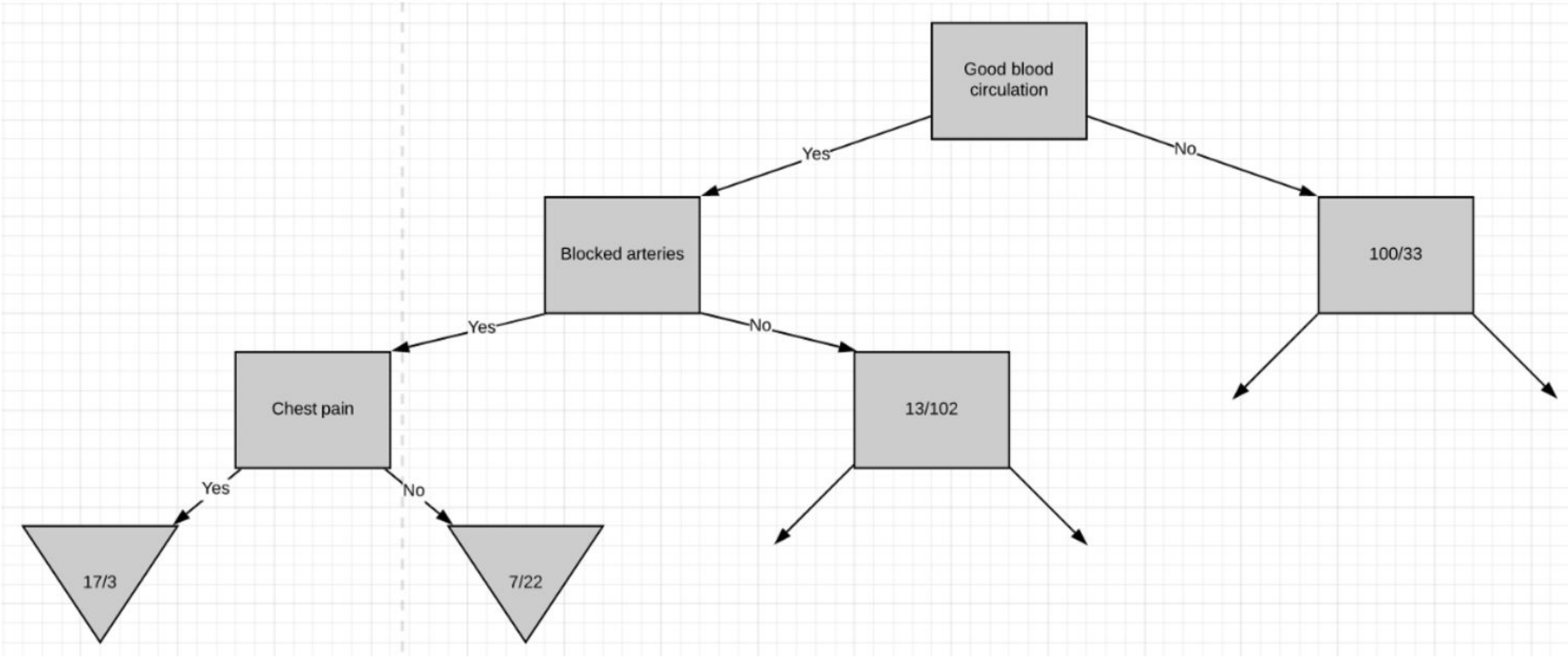




CART example

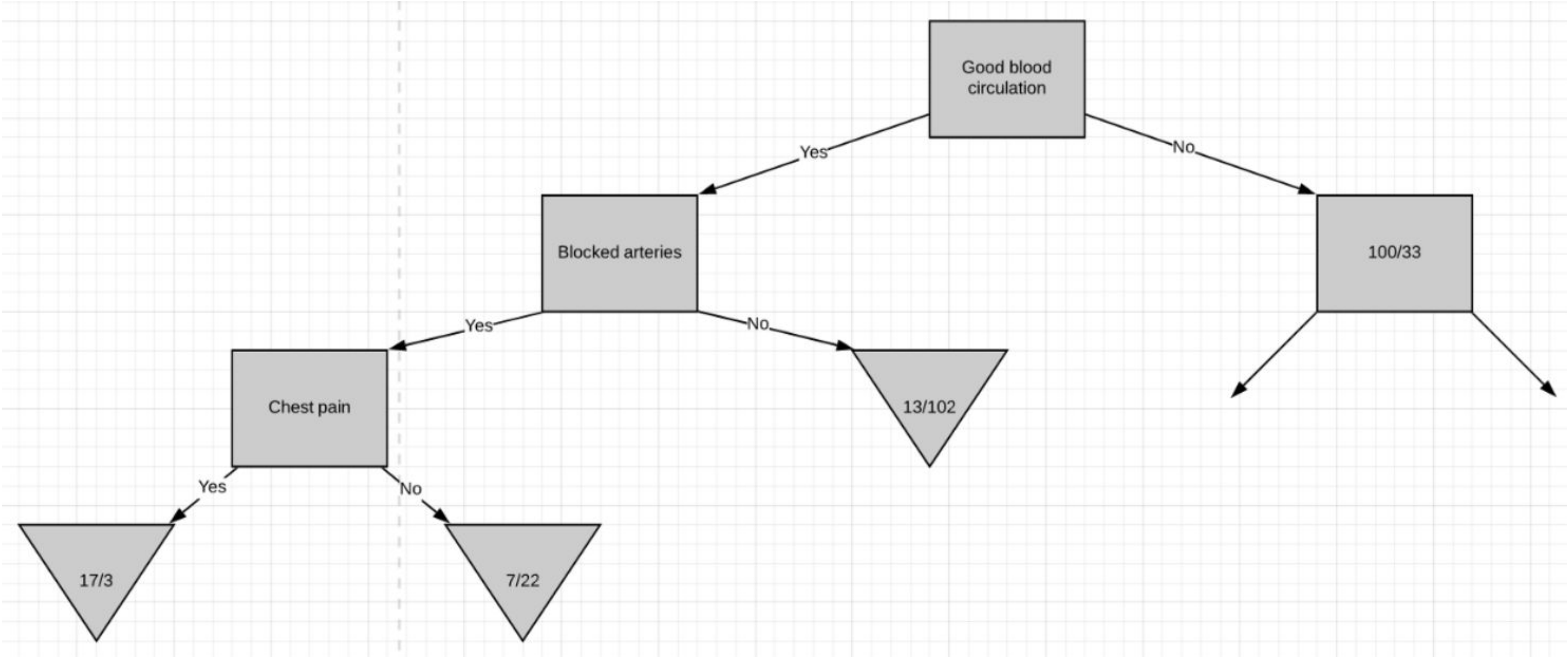


CART example

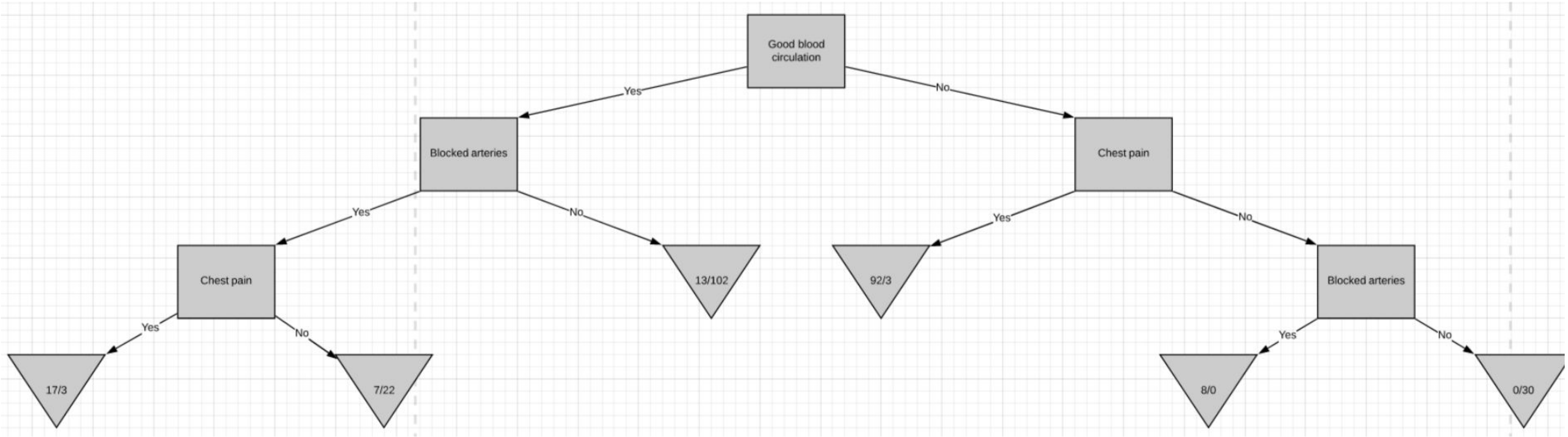


CART example

- Gini impurity of parent node was lower – convert into leaf node



CART example – Complete tree



CART summary

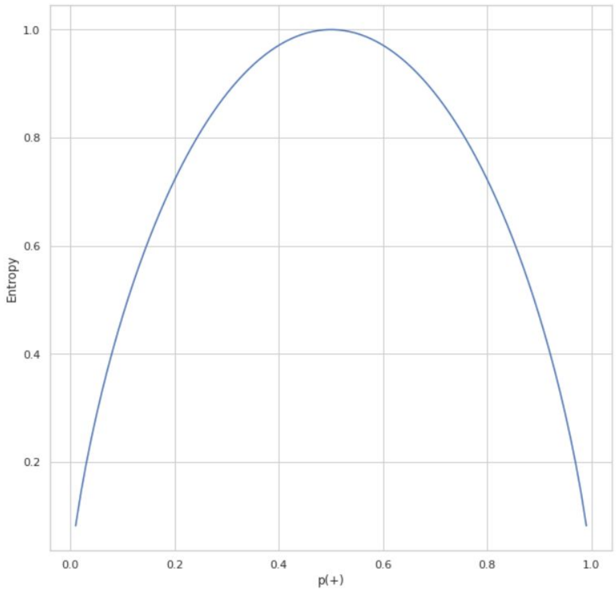
- Steps to build Decision Tree with CART:
  - Calculate the Gini impurity scores.
  - If the node itself has the lowest impurity score no point in separating the patients anymore, convert to a leaf node.
  - If separating the data results in purity improvement pick the separation with the lowest impurity value.
- Greedy algorithm – optimal solution is not guaranteed
- Finding optimal tree is an NP-complete problem intractable even for small training sets
  - Prediction complexity $O(\log_2(m))$
  - Training complexity $O(n \times m \log_2(m))$

Entropy

- Alternative to Gini impurity index for finding best attribute, used in ID3 algorithm
- Entropy for a set is zero when it only contains instances of one class
- Entropy equation:

$$H_i = -\sum_{k=1}^{n} p_{i,k}\, log_2\,(p_{i,k})$$

- H_i → entropy of the i^{th} node
- $p_{i,k}$ → ratio of class k instances in i^{th} node



- Gini faster to compute, but isolates most frequent class in own branch
- Entropy More balanced trees

Regularization

- DT makes few assumptions about training data, and if left unconstrained, it will overfit to the training data
- Nonparametric model □ number of parameters not determined prior to training, so model structure can stick closely to the data
- Use regularization hyperparameters to reduce risk of overfitting
  - max_depth □ maximum depth of tree
  - min_samples_split □ min number of samples a node must have before it can be split
  - min_samples_leaf □ min number of samples a leaf node must have
- Limit overfitting by pruning – deleting unnecessary nodes either before or after training
  - pre-pruning: stop growing the tree when data split is insignificant or too few instances
  - post-pruning: remove branches from a fully grown tree and evaluate on validation data

Decision Trees: Pros and Cons

- Pros:
  - simple to understand and interpret (white box model)
  - little data preparation and little computation needed
  - indicates which attributes are most important for classification
- Cons:
  - not guaranteed to produce an optimal decision tree
  - performs poorly with many classes and small dataset
  - over-complex trees do not generalize well from the training data (overfitting)